The Mansion

Problem set

Written by Claude (Anthropic), 2026-10-22, from the Mansion's statements and scenes. The ideas are dr. kraemer's; the words are the models'.

A value marked † is supplied: chosen so the house works, with the reason somewhere below the mansion. The thirteen problems are the mansion. Appendices a, 2 and γ go further in.

The way in

The inquiry has a place and an hour. Left unspecified: who should arrive, and who will depart.

A visitor arrives to find out how it was all done. The door is unlatched. A loose question sheet lies beside a small blank book, opened to a page half-full of notes.

In the dining room the roast goes cold under its cover. A strip of paper moves beneath a pen. Above it swings a knife.

At the silver pivot, a powdered clown in damask is partway into a curtsy. The flowers on one repaired cuff face inward; every other spray faces out. The curtsy does not deepen.

A trim silver canteen with an enamel juggler on the lid stands on the sideboard. The lock has been polished more recently than the lid.

The visitor watches the knife cycle across and back. On the paper, the curve cycles with it.

Dining room

Problem 1: The knife

A knife is modeled as a point mass $m = 1.0\,\mathrm{kg}$ on a massless rod of length $\ell = 1.0\,\mathrm{m}$, pivoted to swing freely in a vertical plane. It gets released from rest at $\theta_0 = 0.10\,\mathrm{rad}$ from the downward vertical. Use $g = 10\,\mathrm{m/s^2}$ and the small-angle approximation $1 - \cos\theta \approx \theta^2/2$.

1-a Find the gravitational potential energy of the knife at release, relative to its lowest point.

1-b Find its greatest speed.

1-c For small oscillations the angular frequency is $\omega_0 = \sqrt{g/\ell}$, and the period T satisfies $\omega_0 T = 2\pi$. Find T.

1-d How does T change if the mass is doubled?

The pen begins the curve again. The lid lifts.

Kitchen

The line on the paper goes on behind the wall. Ahead, something breathes in steam. By the range, a toolbox in scorched canvas.

Problem 2: The engine and the flywheel

A small steam engine takes in $15\,\mathrm{J}$ of heat and does $3.0\,\mathrm{J}$ of work in each cycle, returning to its initial state at the end of each cycle. It completes one cycle per second.

2-a Find the heat rejected per cycle, the efficiency and the average power output.

2-b The engine's 3.0 W of net power, taken as constant, drives a flywheel of moment of inertia $I = 0.50\,\mathrm{kg\,m^2}$, initially turning at 4.0 rad/s, for 4.0 s. Find the final angular speed.

2-c The power delivered by a torque τ to a wheel turning at angular speed ω is $P = \tau\omega$. Find the driving torque at the start and at the end of the 4.0 s.

Problem 3: The governor and the spool

A centrifugal governor has two 1.0 kg balls, each on a massless 0.50 m arm hinged on the axis of a vertical shaft. The shaft turns steadily at 5.0 rad/s†.

3-a Find the angle each arm makes with the vertical, and how far each ball has risen above its hanging position.

3-b Below what angular speed would the arms hang straight down?

3-c A spool rests on a level counter. It has mass 2.0 kg, outer radius $R = 0.20\,\mathrm{m}$, hub radius $r = 0.12\,\mathrm{m}$, and moment of inertia $0.040\,\mathrm{kg\,m^2}$† about its axle. A light cord wound on the hub leaves from the bottom of the hub and is pulled horizontally with a constant 6.0 N. The spool rolls without slipping. Find its acceleration and the direction it rolls.

3-d If the cord is instead pulled at an angle θ above horizontal, for what θ does the spool stay at rest?

Library

Raise the spool's cord past that angle and, still pulled forward, the spool rolls back. It knocks a catch, and the ladder is free to slide. The first question catches the ladder mid-slide; the rest start it again from rest. On the reading table, a velvet kit.

Problem 4: The ladder

A uniform ladder of mass 8.0 kg and length 3.0 m rests with its foot 1.8 m from a frictionless wall and its top 2.4 m above a frictionless floor. Its moment of inertia about its center is $mL^2/12$. Use $g = 10\,\mathrm{m/s^2}$.

4-a The foot is pulled away from the wall at 0.40 m/s while the top stays against the wall. Find the speed of the top at that instant.

4-b The ladder is released from rest in its original position. It stays in contact with the wall until its top is 1.60 m† above the floor. Find the gravitational potential energy released by then.

4-c While both ends touch, the center moves at speed $L\omega/2$, where ω is the ladder's angular speed. Find how that energy divides between the center's translation and the rotation about the center, and the speed of the center when the top leaves the wall.

Billiard room

After the ladder has left the wall, its fall trips the sear on a cocked cue. The spring pays for the shot. The ladder only lets it go. The chalk is on your cuff. On the sideboard, a clockwork diorama under a striped dome. It has a door in this room, the conservatory and the ballroom.

Problem 5: The cue ball

A uniform solid ball of mass 1.0 kg and radius $R = 0.050$ m (moment of inertia $2mR^2/5$) is at rest on a table. A cue gives it a short horizontal impulse of 2.0 N s at height H above the table. Neglect friction during the impulse.

5-a Find the speed of the ball's center just afterward.

5-b Find H if the ball rolls without slipping immediately.

5-c Find its angular speed and its translational and rotational kinetic energies.

Conservatory

The cue ball rolls free, then nudges a flap. A second ball, held a little way up the basin, is let go. Its own height pays for its motion. The striped dome's second door is here.

Problem 6: The basin

A small uniform solid ball rolls without slipping back and forth across the bottom of a fixed curved basin. Its center moves on a circular arc of radius $a = \frac{5}{7}\,\mathrm{m}^{\dagger}$. For small oscillations its period is $T = 2\pi\sqrt{7a/(5g)}$. Use $g = 10\,\mathrm{m/s^2}$.

6-a Find T, and compare it with the period of the knife in problem 1.

6-b A frictionless bead sliding on the same arc has period $2\pi\sqrt{a/g}$. Find it.

6-c Explain why the rolling ball takes longer.

Ballroom

The ball keeps the knife's time, and a beat that matches draws you on. Their cycles keep time; their starting positions are still theirs to choose. The striped dome's third door is here.

Problem 7: The hoop

A hoop of radius 0.25 m rolls without slipping on a level floor at a steady 8.0 rad/s.

7-a Find the speed of its center.

7-b Find the speeds of the points of the hoop at the top and at the bottom.

7-c Find the speed of a point on the hoop level with the center, at the front.

Problem 8: The ribbon

A 0.50 kg puck slides on a frictionless floor, held by a light ribbon that passes down through a small hole. It moves in a circle of radius 0.60 m at 2.0 m/s. The ribbon is slowly drawn down until the puck again moves in uniform circular motion, now at radius 0.30 m.

8-a Find the puck's new speed and angular speed.

8-b Find the tension in the ribbon in the initial and final circles.

8-c Find the work done by the ribbon.

Study

What supplied that work, and where could it wait? The study keeps the answer ready: two rods held still and a loop held open. A separate little meter has READY marked beside 4 V. Its contact must be held while the needle rises. Bent over the bench, the powdered clown has gloves gone gray at the fingertips, and around one wrist, well above the cuff, a row of old needle holes. Someone took that sleeve in, and wore it. Beside the jars, a doctor's bag in striped oilcloth.

Problem 9: The double pendulum

Two point masses of 1.0 kg hang in series on massless rods each 1.0 m long, swinging in one vertical plane. Use $g = 10\,\mathrm{m/s^2}$. Its small oscillations are combinations of two normal modes†: a slow mode with $\omega_a^2 = 10(2-\sqrt{2})\,\mathrm{s^{-2}}$ ($\omega_a \approx 2.42\,\mathrm{rad/s}$), in which the lower rod's angle is $+\sqrt{2}$ times the upper rod's, and a fast mode with $\omega_b^2 = 10(2+\sqrt{2})\,\mathrm{s^{-2}}$ ($\omega_b \approx 5.84\,\mathrm{rad/s}$), in which it's $-\sqrt{2}$ times. Both angles are measured from the downward vertical, positive toward the same side.

9-a Find the period of each mode.

9-b Show that $\omega_b/\omega_a = 1+\sqrt{2}$. Does a motion mixing both modes ever exactly repeat?

9-c Describe each mode in words: which way does the lower mass move compared with the upper?

Problem 10: The coils and the capacitor

10-a A 6.0 V source is connected through a 20 Ω resistor to an uncharged 0.10 F capacitor. Find the initial current, the time constant, and the current, charge and stored energy when the capacitor reaches 4.0 V.

10-b An ideal LC circuit has $L = 0.10\,\mathrm{H}$ and $C = 1/(L\omega_a^2) \approx 1.71\,\mathrm{F}$†, with ω_a from problem 9. Find its angular frequency $\omega_0 = 1/\sqrt{LC}$, and compare it with ω_a.

10-c A coil fixed to a pendulum rod sits in a magnetic field. With the rod's angle a and the current i both signed, the current exerts a torque $\tau_{\mathrm{em}} = \kappa i$ on the rod, and the rod's motion induces an emf $\varepsilon = -\kappa\dot{a}$ in the circuit. Show that the mechanical power delivered to the rod, $\kappa i\dot{a}$, equals the electrical power the circuit gives up, and that the exchange runs the other way when κi and $\dot{a}$ have opposite signs.

10-d When the slow pendulum mode of problem 9 and the LC circuit are coupled in this way, the combined system has two modes†, at $\frac{2}{3}\omega_0$ and $\frac{3}{2}\omega_0$. Find them, their product, and the ratio of their angular frequencies.

Hall

The rods are left hanging still and the loop open. The hall asks what could announce the release. On the hall stand, a striped hatbox.

Problem 11: The line

A lossless transmission line has inductance 0.50 µH/m and capacitance 0.20 nF/m per unit length. Its wave speed is $u = 1/\sqrt{L'C'}$ and its characteristic impedance is $Z = \sqrt{L'/C'}$. Take $c = 3.0\times10^8\,\mathrm{m/s}$.

11-a Find u and Z.

11-b Two 6.0 V pulses travel in opposite directions and overlap. In each, the current is the voltage divided by Z, in its own direction of travel. Find the total voltage, the total current, and the energy stored per meter, $(C'V^2 + L'I^2)/2$, where they overlap.

11-c Two sinusoidal waves of equal amplitude, at 5.0 MHz and 4.0 MHz†, travel in opposite directions. Their sum has crests that move at $u(f_+ - f_-)/(f_+ + f_-)$ and at $u(f_+ + f_-)/(f_+ - f_-)$†. Find both speeds and compare the faster with c. Can the faster crests carry a signal?

Lounge

A signal goes down the line at u, and two lamps answer it, aimed at your chair from either side. Across the cold hearth, the old joey keeps a hand on the clock key and looks straight at you. They've been looking since before you sat down. The silk at the shoulders has faded round two small thumbprints. On the mantel, a traveling clock case in faded red leather.

Problem 12: The two lights

In a frame S, a light pulse of energy 80 mJ† travels to the right and one of 20 mJ† travels to the left. Take $c = 3.0 \times 10^8$ m/s and $\gamma = 1/\sqrt{1 - v^2/c^2}$.

12-a Find the total energy and the total momentum of the pair in S.

12-b The pair's invariant mass M satisfies $(Mc^2)^2 = E^2 - (pc)^2$. Find Mc^2.

12-c Find the speed v of the frame in which the pair's total momentum is zero, and γ for that speed.

12-d A clock at rest in that frame: how much proper time Δt_0 does it record while 10 s pass in S?

In the chair's frame the opposed pulses have equal energy. The receiver captures them and transfers their 80 mJ to its capacitor. The charging path is disconnected. The loop closes, the rods are released without a kick, and the chair's clock starts.

Problem 13: The chair

A sealed receiver moves through S at the speed found in 12-c. In its rest frame H, time is t_H. Inside is the coupled apparatus of problem 10. It starts with its capacitor holding 80 mJ, the rods vertical and at rest, and no current. After charging, the apparatus is lossless; its only energy stores are the capacitor, the inductor, the rods' kinetic energy and their gravitational potential energy, measured from vertical. Ignore recoil of the support. At $t = t_H = 0$, the receiver's seat is at $x = x_H = 0$. Write its two mode angular frequencies from 10-d as 4ν and 9ν, so $\nu = \omega_0/6$. You see that the charge follows $q = q_0\left[\frac{9}{13}\cos(4\nu t_H) + \frac{4}{13}\cos(9\nu t_H)\right]$†, and the upper rod's angle follows a sum of $\sin(4\nu t_H)$ and $\sin(9\nu t_H)$†.

13-a Find q_0 and the capacitor's initial voltage.

13-b Find the first time $T > 0$ at which the whole initial state returns.

13-c Show that at $T/2$ the current is zero and the rods pass through vertical. Find q/q_0 there and the share of the 80 mJ held by the capacitor. Where is the rest?

13-d How long does T last in S, and how far does the receiver travel in S during it? Using $x_H = \gamma(x - vt)$, find the receiver's position in H throughout.

Appendix a: the cabinet

striped canvas in faded red and cream; brass corners; sawdust and lamp oil; painted stars on every lid; no one else in here; each drawer opens from its room's locked box, and you leave the way you came

Problem a: the kitchen's two 3–4–5s

Door: the kitchen. A toolbox in scorched canvas, patched with the same cloth turned inside out, its lid painted with a ringmaster's star; inside, a little governor on a stand and a spool on a strip of counter.

Use $g = 10\,\mathrm{m/s^2}$.

a-1 A ball hangs on a light arm of length $\ell = 0.50\,\mathrm{m}$, hinged on the axis of a vertical shaft that turns steadily at angular speed ω. Show that the arm settles at an angle θ from the vertical with $\cos\theta = g/(\ell\omega^2)$.

a-2 Find the ω at which the arm's vertical drop and horizontal reach are in the ratio 4 to 3, and how far the ball has risen from hanging straight down.

a-3 A spool of mass m, outer radius R, hub radius r and moment of inertia I about its axle rests on a level counter. A cord wound on the hub leaves from the bottom of the hub and is pulled horizontally with force F. The spool rolls without slipping. Show that its center's acceleration is $a = F(1 - r/R)/(m + I/R^2)$.

a-4 For $m = 2.0\,\mathrm{kg}$ and $R = 0.20\,\mathrm{m}$, find the I that makes $m + I/R^2$ equal $3.0\,\mathrm{kg}$. What simple shape of spool has that I? With $r = 0.12\,\mathrm{m}$ and $F = 6.0\,\mathrm{N}$, find a and the friction force.

a-5 The cord is instead pulled at an angle θ above the horizontal. Show that the spool can stay at rest only when $\cos\theta = r/R$, whatever I is, and find θ.

Problem b: the wall lets go

Door: the library. A gentleman's reading kit in circus colors, its velvet worn to the nap: a folding rule, spectacles, a little ladder leaning on a painted wall.

A uniform ladder of mass m and length L leans on a frictionless wall from a frictionless floor, its top at height h_0. It's released from rest. Use $g = 10\,\mathrm{m/s^2}$.

b-1 While both ends touch, show that the ladder's center stays $L/2$ from the corner where the wall meets the floor.

b-2 While both ends touch, show that the ladder's kinetic energy is $\frac{1}{6}mL^2\omega^2$, where ω is its angular speed.

b-3 The wall can push but can't pull. Show that the top leaves the wall when its height is $2h_0/3$.

b-4 For $L = 3.0\,\mathrm{m}$ and $h_0 = 2.4\,\mathrm{m}$, find that height and the center's speed then.

The back of this drawer opens into the funhouse.

Problem c: turning about the contact

Doors: the conservatory, the billiard room, the ballroom. A clockwork diorama under a striped dome, its base trimmed in baize, taffeta and moss-damp velvet, a strip for each door: a glass marble in a tin basin, a tiny cue, a wire hoop.

A uniform solid ball has mass m, radius R and moment of inertia $\frac{2}{5}mR^2$ about its center. Use $g = 10\,\mathrm{m/s^2}$.

c-1 It rolls without slipping, its center at speed v. Show that its kinetic energy is $\frac{7}{10}mv^2$, $\frac{7}{5}$ of a sliding bead's at the same speed.

c-2 It rolls back and forth in a basin, its center on an arc of radius a. Show that small oscillations have $T = 2\pi\sqrt{7a/(5g)}$, and find the a that matches a simple pendulum 1.0 m long.

c-3 The ball starts at rest on a horizontal table, and a cue gives it a short horizontal impulse at height H above the table. Neglect the table's horizontal impulse during the strike. Show that it rolls at once only if $H = 7R/5$.

c-4 A hoop rolls with its center at speed v. Find the speeds of its top and bottom points. Which point of the hoop is still?

Problem d: the second knife's two motions

Door: the dining room. A silver canteen with an enamel juggler on the lid, lined in damask; inside, under glass, one knife hanging from another.

Two point masses m hang in series on light rods each ℓ long, swinging in one vertical plane. For small angles θ_1 (upper rod) and θ_2 (lower rod) from the downward vertical, positive toward the same side,

$$2\ddot{\theta}_1 + \ddot{\theta}_2 = -2(g/\ell)\,\theta_1, \qquad \ddot{\theta}_1 + \ddot{\theta}_2 = -(g/\ell)\,\theta_2.$$

d-1 In a normal mode both angles go as $\cos\omega t$, with $\theta_2 = +\sqrt{2}\,\theta_1$ or $\theta_2 = -\sqrt{2}\,\theta_1$. Find ω^2 for each.

d-2 With $\ell = 1.0\,\mathrm{m}$ and $g = 10\,\mathrm{m/s^2}$, find both ω and their ratio. Does a motion with both modes ever exactly repeat?

The back of this drawer opens into the funhouse.

Problem e: a can to match the knife

Door: the study. A doctor's bag in striped oilcloth: a capacitor the size of a teacup, a coil, a card of figures, the first of them 22/7.

An LC circuit has $L = 0.10\,\mathrm{H}$.

e-1 Find the C that makes $\omega_0 = 1/\sqrt{LC}$ equal the slow mode of problem d, $\omega_0^2 = 10(2-\sqrt{2})\,\mathrm{s^{-2}}$.

e-2 With that C, how long does one turn of the circuit take, and one turn of the slow mode?

The back of this drawer opens into the funhouse.

Problem f: two crests from two waves

Door: the hall. A hatbox in red and white stripes, rubberized against the wet; inside, a long coil of wire and two tuning forks.

Two waves of equal amplitude A run in opposite directions on a line with wave speed u: $A\cos(k_1x - \omega_1t)$ to the right and $A\cos(k_2x + \omega_2t)$ to the left, with $k = \omega/u$ for each.

f-1 Show that their sum is $2A\cos\big(\frac{1}{2}[(k_1 - k_2)x - (\omega_1 + \omega_2)t]\big)\cos\big(\frac{1}{2}[(k_1 + k_2)x - (\omega_1 - \omega_2)t]\big)$.

f-2 Find the speed of each factor's crests, in terms of u and the two frequencies.

f-3 For 5.0 MHz and 4.0 MHz and $u = 1.0 \times 10^8$ m/s, find both speeds.

The back of this drawer opens into the funhouse.

Problem g: two lights made equal

Door: the lounge. A traveling clock case in faded red leather, lined in quilted silk; inside, two little lamps, one brighter than the other.

Seen from a frame moving at βc in a light pulse's own direction, the pulse's energy is multiplied by $\sqrt{(1-\beta)/(1+\beta)}$; seen from one moving against it, by $\sqrt{(1+\beta)/(1-\beta)}$. A pulse of energy E_1 runs right and one of E_2 runs left, with $E_1 > E_2$.

g-1 Find β for the frame moving right in which the two energies are equal, in terms of $r = E_1/E_2$.

g-2 Show that $\gamma = (r+1)/(2\sqrt{r})$ there.

g-3 For $r = 4$, find β and γ, and show that $r-1$, $2\sqrt{r}$ and $r+1$ make a right triangle.

The back of this drawer opens into the funhouse.

Appendix 2: the funhouse

mirrors on every wall, some flat and some not; a floor that tilts one way and then the other; everything you know comes back at you flipped; no one else in here

Problem 1: the knives, flipped

A long mirror in which the second knife hangs upside down.

The double pendulum of problem d has $2\ddot{\theta}_1 + \ddot{\theta}_2 = -2(g/\ell)\,\theta_1$ and $\ddot{\theta}_1 + \ddot{\theta}_2 = -(g/\ell)\,\theta_2$, with $\ell = 1.0\,\mathrm{m}$ and $g = 10\,\mathrm{m/s^2}$.

1-α In a normal mode $\theta_2 = r\theta_1$ and both angles go as $\cos\omega t$. Use both equations to show that $r^2 = 2$.

1-β Show that the two modes' angular frequencies satisfy $\omega_b/\omega_a = 1+\sqrt{2}$ and $\omega_a/\omega_b = \sqrt{2}-1$, and that $\omega_a^2\omega_b^2 = 2(g/\ell)^2$.

1-γ At what angular rate does the relative phase of the two modes advance? For equal contributions to θ_1 that begin in phase, how long is the interval between successive reinforcements? Does that interval return the whole state?

Trapdoor, the swing side: in the floor under the mirror, and through it the same two rods, full size, on a black iron chassis with a chipped ceramic insulator. They're humming. The trapdoor is open.

Problem 2: the trade, both ways

Two facing mirrors; the can in one is the coil in the other.

The slow mode of the double pendulum acts like one swing a with effective inertia η and stiffness D, so its free angular frequency is $\omega_0 = \sqrt{D/\eta}$. A coil in a magnetic field couples it to an LC circuit with $\kappa > 0$ and the same $\omega_0 = 1/\sqrt{LC}$:

$$\eta\ddot{a} = -Da + \kappa i, \qquad L\ddot{q} = -q/C - \kappa\dot{a}, \qquad i = \dot{q}.$$

2-α Try $a = A\cos\omega t$ and $q = Q\sin\omega t$. Show that $\eta L(\omega_0^2 - \omega^2)^2 = \kappa^2\omega^2$.

2-β Show that the two positive solutions multiply to ω_0^2 and that their gap $\Delta\omega = \omega_{\mathrm{high}} - \omega_{\mathrm{low}}$ satisfies $\Delta\omega\sqrt{\eta L} = \kappa$.

2-γ What κ puts the two modes at $\frac{2}{3}\omega_0$ and $\frac{3}{2}\omega_0$? With $\omega_0 = 2.42\,\mathrm{rad/s}$, find both.

Trapdoor, the circuit side: windings and a capacitor the size of a hatbox, wired into the same chassis, with the chipped insulator again. The trapdoor is open.

Problem 3: two more mirrors

A hall of mirrors that never ends, and a lamp in it that can't decide how bright it is.

3-α For problem f's two distinct positive frequencies, with $f_1 > f_2$, show that the slow and fast crest speeds multiply to u^2.

3-β On a lossless line a wave pattern stores $(C'V^2 + L'I^2)/2$ per meter and carries power VI. Show that $|VI|$ is never more than u times the stored energy per meter. Can the fast crest carry a signal?

3-γ For problem g's lights, show that the two energies multiply to the same number in every frame moving along the line, and check it with 80 mJ and 20 mJ. For a source receding

directly along the line of sight at constant velocity, flashes arrive $s = 2$ times farther apart than their emission interval on the source's clock. Show that $(s + 1/s)/2$ is problem g's γ.

Trapdoor, the timing side: a clock face on the same chassis, one hand, marked in turns, with the chipped insulator under it. The trapdoor is open.

Problem 4: the corner, turned over

A tilted floor, and a mirror at the end that shows the room upside down.

While the ladder of problem b is moving and both ends still touch their supports, its foot is at $(L\cos\varphi, 0)$ and its top at $(0, L\sin\varphi)$, with $0 < \varphi < \pi/2$. The corner where the wall meets the floor is the origin.

4-α The foot moves along the floor and the top along the wall. Show that the point that's still for the instant, where the perpendiculars to the two ends' motions meet, is $(L\cos\varphi, L\sin\varphi)$.

4-β Show that this point is the corner turned a half turn about the ladder's center, so it's always $L/2$ from the center.

4-γ About that point all the motion is turning. Show that the kinetic energy is $\frac{1}{2}m(L^2/3)\omega^2$, and find the energy per kilogram when the top leaves the wall, in problem b's numbers.

In the far glass, the nine costumes hang empty and inside out. The long coat's lining stops at an ordinary person's height. Through the worn knees of the velvet diamonds, the pattern continues in another color.

Each costume is missing one stitch. It's the same stitch.

They were all cut for one wearer.

Appendix γ: under the stage

a low room under the stage; one machine on a black iron chassis, bigger than it looked through the trapdoors; the chipped ceramic insulator; a hum not quite loud enough to cover the whispers; stagehand blacks on a nail, still warm; no one else down here

22/7 has been painted on the chassis, rubbed off, and painted again. The first drawing is a circle and a square. The toymaker had meant to square the circle and be done.

The blacks on the nail are patched with quilted silk. Its flattened hollows have kept the shape of somebody sitting.

Problem α: the machine, all of it

Any of the three trapdoors comes down here.

Take the coupled machine of appendix 2, problem 2, with its tones at $\frac{2}{3}\omega_0$ and $\frac{3}{2}\omega_0$, written 4ν and 9ν, so $\nu = \omega_0/6$. Scale the swing and the charge as $X = \sqrt{\eta}\,a$ and $Y = \sqrt{L}\,q$, so that

$$\ddot{X} = -\omega_0^2 X + \Delta\omega\,\dot{Y}, \qquad \ddot{Y} = -\omega_0^2 Y - \Delta\omega\,\dot{X},$$

with the gap $\Delta\omega = 5\nu$. It starts with $Y = Y_0 \neq 0$, $X = 0$, and $\dot{X} = \dot{Y} = 0$.

α-1 Show that the slow mode has $X = -B_1 \sin 4\nu t$ and $Y = B_1 \cos 4\nu t$, and the fast mode $X = B_2 \sin 9\nu t$ and $Y = B_2 \cos 9\nu t$.

α-2 Fit the start and show that $B_1 = \frac{9}{13}Y_0$ and $B_2 = \frac{4}{13}Y_0$.

α-3 Find the first time the whole state returns, and how the energy is shared between the swing's motion and the capacitor halfway there.

α-4 Using $\omega_0^2 = 10(2-\sqrt{2})\,\mathrm{s}^{-2}$ and 80 mJ in the machine, how long is the return?

From a corner under the frame: "Here I zam."

Its twin meets its eyes.

"Zlam."

The hum covers them again.

Problem β: TEST, then RUN

On the chassis: a switch with two positions, TEST and RUN, and a small drive arm linked to the mechanical coordinate a.

In TEST the electrical loop is open, so $i = 0$. The drive acts only on the mechanical coordinate a; the other mechanical mode is absent. Thus $X = \sqrt{\eta}\,a$ obeys $\ddot{X} = -\omega_0^2 X$ between taps. The drive gives N short impulses, each adding the same positive increment to $\dot{X}$. Start from rest, and let E_1 be the energy one tap gives this oscillator from rest.

β-1 If successive taps are separated by one complete period of this oscillator, show that after N taps its energy is $N^2 E_1$, and that the k-th tap does $(2k-1)E_1$ of work.

β-2 If the free phase advances between taps are independent and uniformly distributed over a full cycle, show that the expected energy after N taps is NE_1.

β-3 Compare the two after 100 taps.

β-4 In RUN the drive is disconnected and the machine is lossless. Does its energy change when its two tones line up at the return?